

EFREI – Épreuve E6

Chaîne CI/CD conteneurisée

Module : Projet Intégration et Déploiement Continu

Rapport de rendu

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Mention : **Anonyme**

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Prompt initial

Refait le même squelette que E5 mais cette fois pour E6 s'il te plaît.

1 Partie 1 – Git et gestion de versions (20 points)

1.1 Exercice 1 – Initialisation et structuration du dépôt (8 points)

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

1.a Dépôt Git

```
abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend
```

FIGURE 1 – Dépôt Git

```
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git add .
git commit -m "feat: initialisation du projet VitalSync avec fichiers de départ"
On branch main
nothing to commit, working tree clean
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % gh repo create vitalsync --public --source=. --push
✓ Created repository Amakran2003/vitalsync on github.com
https://github.com/Amakran2003/vitalsync
✓ Added remote https://github.com/Amakran2003/vitalsync.git
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 10 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (14/14), 166.64 KiB | 33.33 MiB/s, done.
Total 14 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Amakran2003/vitalsync.git
 * [new branch] HEAD -> main
branch 'main' set up to track 'origin/main'.
✓ Pushed commits to https://github.com/Amakran2003/vitalsync.git

A new release of gh is available: 2.83.1 → 2.89.0
To upgrade, run: brew upgrade gh
https://github.com/cli/cli/releases/tag/v2.89.0
```

FIGURE 2 – Premier commit

Justification du choix GitHub / GitLab : J'ai choisi GitHub car il est plus facile à utiliser et à déployer.

1.b Stratégie de branches Gitflow

```
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git checkout -b develop
git push -u origin develop
Switched to a new branch 'develop'
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/Amakran2003/vitalsync/pull/new/develop
remote:
To https://github.com/Amakran2003/vitalsync.git
 * [new branch]      develop -> develop
branch 'develop' set up to track 'origin/develop'.
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git checkout -b feature/docker-setup
git push -u origin feature/docker-setup
Switched to a new branch 'feature/docker-setup'
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/docker-setup' on GitHub by visiting:
remote:   https://github.com/Amakran2003/vitalsync/pull/new/feature/docker-setup
remote:
To https://github.com/Amakran2003/vitalsync.git
 * [new branch]      feature/docker-setup -> feature/docker-setup
branch 'feature/docker-setup' set up to track 'origin/feature/docker-setup'.
```

FIGURE 3 – Liste des branches créées

1.c Fichier .gitignore

.gitignore

```
1 node_modules/
2 .env
3 dist/
4 *.log
5 .DS_Store
6
```

Justification des entrées :

- Il ne faut surtout pas inclure les fichiers de `node_modules`, `.env`, `dist`, `*.log` et `.DS_Store` dans le dépôt Git.

1.d Protection de la branche main

```
abderazagmakran@Abderazags-MacBook-Air vitalsync % gh api repos/{owner}/vitalsync/rulesets \
--method POST \
--input - <<'EOF'
{
  "name": "Protect main",
  "target": "branch",
  "enforcement": "active",
  "conditions": {
    "ref_name": {
      "include": ["refs/heads/main"],
      "exclude": []
    }
  },
  "rules": [
    {
      "type": "pull_request",
      "parameters": {
        "required_approving_review_count": 1,
        "dismiss_stale_reviews_on_push": false,
        "require_code_owner_review": false,
        "require_last_push_approval": false,
        "required_review_thread_resolution": false
      }
    }
  ]
}
EOF
{
  "id": 14593308,
  "name": "Protect main",
  "target": "branch",
  "source_type": "Repository",
  "source": "Amakran2003/vitalsync",
  "enforcement": "active",
  "conditions": {
    "ref_name": {
      "exclude": [],
      "include": [
        "refs/heads/main"
      ]
    }
  },
  "rules": [
    {
      "type": "pull_request",
      "parameters": {
        "required_approving_review_count": 1,
        "dismiss_stale_reviews_on_push": false,
        "required_reviewers": [],
        "require_code_owner_review": false,
        "require_last_push_approval": false,
        "required_review_thread_resolution": false,
        "allowed_merge_methods": [
          "merge",
          "squash",
          "rebase"
        ]
      }
    }
  ]
},
  "node_id": "RRS_1ACqUwVw3NpdG9yc5HbJxxzgDcrRw",
  "created_at": "2026-04-01T14:12:30.202+02:00",
  "updated_at": "2026-04-01T14:12:30.213+02:00",
  "bypass_actors": [],
  "current_user_can_bypass": "never",
  "links": {
    "self": {
      "href": "https://api.github.com/repos/Amakran2003/vitalsync/rulesets/14593308"
    },
    "html": {
      "href": "https://github.com/Amakran2003/vitalsync/rules/14593308"
    }
  }
}
```

FIGURE 4 – Règles de protection de la branche main

1.2 Exercice 2 – Workflow Git et résolution de conflits (12 points)

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

2.a Branches parallèles et conflit

Version sur feature/add-endpoint :

backend/server.js – branche feature/add-endpoint

```
1  const express = require("express");
2  const app = express();
3
4  app.get("/health", (req, res) => {
5    res.json({ status: "ok", timestamp: new Date() });
6  });
7
8  app.get("/api/users", (req, res) => {
9    res.json([ { id: 1, name: "Alice" } ]);
10 });
11
12 app.get("/api/users", (req, res) => {
13   res.json([ { id: 1, name: "John Doe" } ]);
14 });
```

Version sur feature/update-health :

backend/server.js – branche feature/update-health

```
1  const express = require("express");
2  const app = express();
3
4  app.get("/health", (req, res) => {
5    res.json({ status: "ok", version: "1.0.0", timestamp: new Date() });
6  });
7
8  app.get("/api/users", (req, res) => {
9    res.json([ { id: 1, name: "Alice" } ]);
10 });
```

Extrait messages du conflit : Extrait du conflit généré (marqueurs Git) :

```
Auto-merging backend/server.js
CONFLICT (content): Merge conflict in backend/server.js
Automatic merge failed; fix conflicts and then commit the result.
```

FIGURE 5 – backend/server.js – conflit messages

```

const express = require("express");
const app = express();
app.get("/health", (req, res) => {
  res.json({ status: "ok", timestamp: new Date() });
});
app.get("/api/activities", (req, res) => {
  res.json([]);
});
Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes | Resolve with AI
<<<<<< HEAD (Current Change)
app.get("/api/users", (req, res) => {
  res.json([{ id: 1, name: "John Doe" }]);
}
=====
app.get("/api/health", (req, res) => {
  res.json({ id: 1, name: "Alice" });
}
>>>>>> feature/update-health (Incoming Change)
});
app.listen(3000, () => console.log("VitalSync API on :3000"));

```

FIGURE 6 – backend/server.js – conflit

2.b Résolution du conflit

backend/server.js – après résolution du conflit

```

1  const express = require("express");
2  const app = express();
3
4  app.get("/health", (req, res) => {
5    res.json({ status: "ok", version: "1.0.0", timestamp: new Date() });
6  });
7
8  app.get("/api/users", (req, res) => {
9    res.json([{ id: 1, name: "Alice" }]);
10 });
11
12 app.get("/api/users", (req, res) => {
13   res.json([{ id: 1, name: "John Doe" }]);
14 });

```

2.c Merge vs Rebase

La différence entre Merge et Rebase est que Merge crée un nouveau commit qui contient les modifications des deux branches, tandis que Rebase applique les modifications de la branche cible sur la branche source.

2.d Conventional Commits

Justification de la convention utilisée : [À compléter]


```

abderrazqmakran@Abderrazaqs-MacBook-Air vitalsync % git add backend/server.js
abderrazqmakran@Abderrazaqs-MacBook-Air vitalsync % git commit -m "fix: resolve merge conflict in backend/server.js"
[develop 8e209ff] fix: resolve merge conflict in backend/server.js
abderrazqmakran@Abderrazaqs-MacBook-Air vitalsync % git status
On branch develop
Your branch is ahead of 'origin/develop' by 2 commits.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   rapport-e6-vitalsync.pdf
        modified:   rapport-e6-vitalsync.tex

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        img/.png
        img/docker-build-backend.png
        img/docker-compose-up.png
        img/docker-images-backend.png
        img/docker-ps.png
        img/git-branches.png
        img/git-conflict-diff.png
        img/git-conflict-marqueurs.png
        img/git-first-commit.png
        img/git-log-conventional.png
        img/git-merge-reussi.png
        img/git-protection.png
        img/git-repo.png
        img/gitignore.png
        img/pipeline-build-docker.png
        img/pipeline-deploy-staging.png
        img/pipeline-etapes-detail.png
        img/pipeline-lint-tests.png
        img/pipeline-reussie.png
        img/pipeline-triggers.png
        img/schema-architecture.png
        img/secrets-config.png

no changes added to commit (use "git add" and/or "git commit -a")

```

FIGURE 7 – Commit de merge réussi

```

• abderrazqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazqmakran/Documents/examen/vitalsync
• abderrazqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazqmakran/Documents/examen/vitalsync/.git/
• abderrazqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazqmakran/Documents/examen/vitalsync/.git/
• abderrazqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend

```

FIGURE 8 – Historique Git montrant au moins 5 commits Conventional Commits

2 Partie 2 – Conteneurisation Docker (25 points)

2.1 Exercice 3 – Dockerfile du back-end (10 points)

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

3.a Dockerfile back-end (multi-stage)

```
backend/Dockerfile
1 # A compléter
```

Explication du multi-stage build : [À compléter – pourquoi 2 stages, intérêt de chacun]
Justification de l’image de base : [À compléter – pourquoi node :20-alpine et pas node :20]

3.b Commentaire explicatif (en tête du Dockerfile)

[Déjà intégré dans le Dockerfile ci-dessus]

3.c Fichier .dockerignore

```
.dockerignore
1 # A compléter
```

Justification des entrées :

chaque entrée

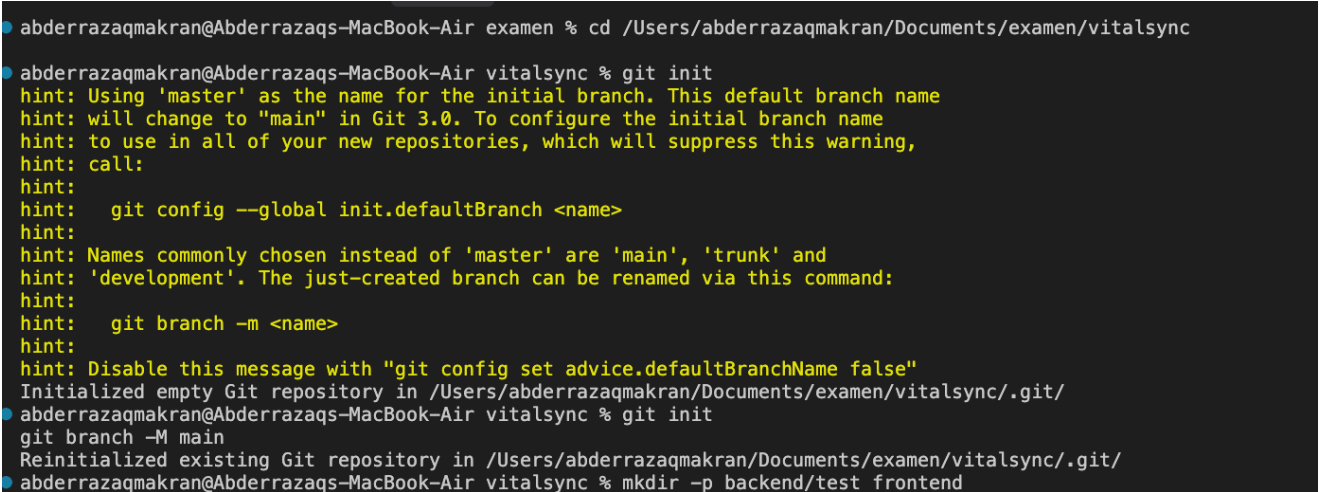


FIGURE 9 – Build réussi du Dockerfile back-end montrant les étapes

2.2 Exercice 4 – Dockerfile du front-end (5 points)

Prompt

--

9

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

4.a Dockerfile front-end

frontend/Dockerfile

```
1 # A compléter
```

Justification de l'image Nginx utilisée : [À compléter]

4.b Fichier nginx.conf

frontend/nginx.conf

```
1 # A compléter
```

Explication du rôle du proxy_pass : [À compléter – pourquoi les requêtes /api/* sont redirigées vers le back-end]

2.3 Exercice 5 – Docker Compose (10 points)

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

5.a Fichier docker-compose.yml

docker-compose.yml

```
1 # A compléter
```

5.b Réseau personnalisé (bridge)

[À compléter – pourquoi un bridge dédié, avantage par rapport au réseau par défaut]

5.c Volume persistant

[À compléter – qu'est-ce qui se passe si on fait `docker-compose down` sans volume]

5.d Variables d'environnement

.env.example

```
1 # A compléter
```

Justification de chaque variable :

À compléter

5.e Vérification

```

abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend

```

FIGURE 10 – docker images montrant la taille de l'image produite

```

abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend

```

FIGURE 11 – docker-compose up montrant le démarrage des 3 services

```

abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend

```

FIGURE 12 – docker ps montrant les 3 conteneurs actifs avec leurs ports

3 Partie 3 – Pipeline CI/CD (30 points)

3.1 Exercice 6 – Configuration de la pipeline (20 points)

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

Justification du choix **GitHub Actions** / **GitLab CI** : [À compléter]

6.a Fichier de pipeline

.github/workflows/ci.yml

1 # A compléter

Étape 1 – Lint & Tests (6 pts)

Explication : [À compléter – que fait cette étape, pourquoi]

```
abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend
```

FIGURE 13 – Logs du lint et des tests

Étape 2 – Build Docker (6 pts)

Explication : [À compléter – que fait cette étape, pourquoi]

Justification du choix du registry : [À compléter – GHCR, Docker Hub ou GitLab Registry]

Explication du tag par SHA de commit : [À compléter – pourquoi c'est préférable à un tag à latest
z]

Étape 3 – Déploiement staging (8 pts)

Explication : [À compléter – que fait cette étape, pourquoi]

Explication du health check : [À compléter – comment il fonctionne, pourquoi il fait échouer la
pipeline]

```

• abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend

```

FIGURE 14 – Logs du build Docker

```

• abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend

```

FIGURE 15 – Logs du déploiement staging

```

• abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
• abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend

```

FIGURE 16 – Onglet Actions/Pipelines montrant une exécution réussie

```
abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend
```

FIGURE 17 – Détail de chaque étape de la pipeline

Pipeline réussie

3.2 Exercice 7 – Gestion des secrets et déclencheurs (10 points)

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

7.a Secrets configurés

```
abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend
```

FIGURE 18 – Page de configuration des secrets (valeurs masquées)

Liste des secrets et leur rôle :

ès au registry

7.b Déclencheurs

```
abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name> 14
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
```

7.c Risques du stockage de secrets en clair

[À compléter – au moins 2 risques concrets]

4 Partie 4 – Orchestration et supervision (15 points)

4.1 Exercice 8 – Manifestes Kubernetes (10 points)

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

8.a Deployment back-end

k8s/deployment.yaml

```
1 # A compléter
```

Explication du rôle d'un Deployment : [À compléter]

Justification du choix de 2 réplicas : [À compléter – pourquoi pas 1, pourquoi pas 3]

Explication de la liveness probe : [À compléter – à quoi elle sert, que se passe-t-il si elle échoue]

8.b Service ClusterIP

k8s/service.yaml

```
1 # A compléter
```

Explication du rôle d'un Service : [À compléter]

8.c Ingress / NodePort / LoadBalancer

k8s/ingress.yaml

```
1 # A compléter
```

Explication du choix entre Ingress et NodePort/LoadBalancer : [À compléter]

8.d Secret Kubernetes

k8s/secret.yaml

```
1 # A compléter
```

Explication de l'injection du Secret dans le pod : [À compléter – envFrom ou env.valueFrom]

4.2 Exercice 9 – Réflexion sur la supervision (5 points)

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

9.a Supervision en production

Outils choisis :

on / alerting

Métriques surveillées :

e justification

Justification du choix des outils : [À compléter]

9.b Self-healing Kubernetes

[À compléter – rôle du ReplicaSet, détection de panne, recreation du pod]

5 Partie 5 – Documentation (10 points)

5.1 Exercice 10 – README et documentation technique

Prompt

Prompt utilisé	Fournisseur	Modèle
À compléter	[À compléter]	[À compléter]

10.a Fichier README.md

README.md

```
1 # A compléter
```

10.b Schéma d'architecture

```
abderrazaqmakran@Abderrazaqs-MacBook-Air examen % cd /Users/abderrazaqmakran/Documents/examen/vitalsync
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % git init
git branch -M main
Reinitialized existing Git repository in /Users/abderrazaqmakran/Documents/examen/vitalsync/.git/
abderrazaqmakran@Abderrazaqs-MacBook-Air vitalsync % mkdir -p backend/test frontend
```

FIGURE 20 – Schéma d'architecture du projet VitalSync

Justification des choix techniques : [À compléter]

Checklist finale de rendu

- Page de garde présente (anonyme)
- Sommaire présent
- Parties 1 à 5 complétées
- Captures d'écran intégrées dans le PDF
- Fichiers de configuration intégrés dans le PDF
- Justification explicite de chaque choix
- Transparence IA documentée
- Lien vers le dépôt Git inclus